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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Chili Raccoon 7
Selected Domain	Education Technology (EdTech) & Workforce Development
Selected SDG Track (1–17)	SDG 4 – Quality Education
Project / Solution Name	FutureReady
Name (Leader)	Shiu Jun Ming
Email & Phone (Leader)	shiujm2007@gmail.com +60122968377
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia faces a growing skills mismatch between academic qualifications and industry requirements. Many students, fresh graduates, and job seekers struggle to identify the skills needed for their desired careers. While job portals provide vacancy listings, they rarely offer personalized guidance on how candidates can bridge their skill gaps and improve employability.

As industries rapidly adopt digital technologies and artificial intelligence, required skill sets evolve faster than traditional career-planning methods can accommodate. Consequently, individuals often spend significant time searching across multiple platforms to understand career requirements, evaluate their readiness, and identify relevant learning opportunities. This lack of visibility can lead to poor career decisions, prolonged job-search periods, and underemployment.

The challenge particularly affects university students and early-career professionals who have limited industry exposure. Employers are also impacted because applications frequently come from candidates whose skills do not align with job requirements, resulting in inefficient recruitment processes.

This problem has both educational and economic implications. Without access to personalized career guidance and skill-development pathways, individuals may struggle to remain competitive in an increasingly technology-driven workforce. Addressing this issue aligns with the global objective of preparing learners with relevant skills for employment and lifelong learning.

Our team seeks to address this gap through an AI-powered workforce readiness platform that helps users understand their current capabilities, identify missing skills, and receive targeted recommendations that improve career preparedness.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FutureReady Workforce is an AI-powered career readiness platform designed to help users evaluate, improve, and showcase skills required for modern employment.

The platform enables users to create a professional profile and upload their resume or CV. Using artificial intelligence and natural language processing techniques, the system extracts relevant skills and maps them against selected career pathways. Users are then presented with a personalized skills profile and can review, validate, and assign proficiency levels to each detected skill.

The platform performs skill-gap analysis by comparing a user's existing competencies with industry-required skills for targeted career roles. Based on these gaps, FutureReady Workforce generates personalized learning recommendations and development pathways that help users become more employment-ready.

In addition, users can explore career opportunities aligned with their strengths and identify specific competencies required to qualify for desired positions. By providing actionable recommendations instead of simply listing job vacancies, the platform transforms career planning into a structured, data-driven process.

The solution leverages knowledge acquired during SDC Phase 1 workshops, including:

- Generative AI APIs for skill extraction and recommendation generation.
- Database architecture for storing user profiles, skills, and career pathways.
- Cloud hosting and deployment services for scalable web application delivery.
- Modern web technologies for an accessible and user-friendly interface.

By combining AI-powered analysis with practical career guidance, FutureReady Workforce empowers students and job seekers to make informed decisions, continuously improve their employability, and better prepare for future workforce demands.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend

- React
- TypeScript
- Tailwind CSS
- Vite 7

Backend

- Node.js
- Express.js
- REST API Services

CV & Career Analysis

- PDF/DOCX text extraction and Tesseract.js OCR for supported images
- Users review detected skills and confirm proficiency levels
- Career matching and skill-gap analysis use bundled role/skill references
- Qwen3-4B explanations through RunPod after user consent

Database

- Browser localStorage; no active production database

Deployment

- Replit hosts the website and backend
- RunPod GPU Pod supports the Qwen3-4B explanation service

External API & Sources

- OpenDOSM API and Google Fonts API
- DOSM reports/workbooks and Ministry of Digital announcements webpage (webpage, not API)

Sponsor-Provided Technology

Yes.

The project utilizes Generative AI technologies showcased during the SDC workshops to perform resume skill extraction, skill classification, personalized recommendations, and career readiness analysis. Cloud deployment knowledge obtained from the workshops is also used to host and deliver the web application prototype.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team adopted the design-thinking methodology to ensure that FutureReady Workforce addresses genuine workforce challenges while remaining practical to implement.

Empathize: We identified common difficulties faced by students and job seekers, including uncertainty about career requirements, lack of awareness of skill gaps, and challenges in finding relevant learning pathways.

Define: We framed the core problem as the absence of personalized, data-driven guidance for career preparation and workforce readiness.

Ideate: Multiple solution concepts were explored, including job recommendation systems and learning platforms. We selected an AI-powered workforce readiness platform because it directly connects skills assessment, career planning, and learning recommendations.

Prototype: We developed a functional web-based prototype featuring profile management, CV upload, AI-powered skill extraction, skill validation, and career guidance capabilities.

Test: The solution was iteratively evaluated using sample resumes and career datasets to verify the accuracy of skill extraction and recommendation workflows.

For deployment, the system is built using scalable web technologies and cloud-hosted infrastructure. This approach enables rapid testing, user feedback collection, and future expansion during Phase 2 while maintaining a realistic pathway toward real-world adoption.

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4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 4.4** By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs, and entrepreneurship.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Based on preliminary observations, many students and job seekers lack visibility into the skills required for their intended career paths and do not receive personalized recommendations for closing identified skill gaps. **Baseline Metric:** By the age of 15, 39% of students are unclear about their career expectations, double the proportion from less than a decade ago. One in three students also said that school had not taught them things which could be useful in a job. Source: https://www.oecd.org/en/about/news/press-releases/2025/05/the-state-of-global-teenage-career-preparation.html?adestraproject=What%27s%20New%20at%20OECD&utm_campaign=What%27s%20New%20-%20June%202025&utm_content=teenage-career-image&utm_term=com&utm_medium=email&utm_source=Adestra#1
Core Measurable KPI	**Career Readiness Improvement Rate (%)**

Metric Category	Your Response
The specific unit you will track *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Measured as:(Number of identified skills gaps addressed after recommendations ÷ Total identified skills gaps) × 100 Additional supporting indicators: ● Number of skills successfully extracted from uploaded CVs ● Profile completion rate ● Number of personalized learning recommendations generated ● Percentage of users receiving career pathway guidance
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	● 80% of users successfully generate a personalized skills profile. ● 70% of users identify at least one critical skills gap. ● 75% of users receive personalized learning recommendations. ● 30% improvement in workforce readiness assessment scores after platform usage.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

/	All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
/	The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
/	The proposed solution targets a functional, deployed application rather than a theoretical business case study.
/	All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code / dataset / model)	License / Terms
TypeScript 5.9; JavaScript; Python utility scripts	Code	TypeScript: Apache-2.0; JavaScript: ECMAScript standard; Python: PSF License
React 19	Code	MIT
React DOM 19	Code	MIT
Wouter 3	Code	MIT
Tailwind CSS 4	Code	MIT
Radix UI	Code	MIT; individual package licences apply
Express 5	Code	MIT
Node.js 20	Code	MIT
Vite 7	Code	MIT
Zod	Code	MIT
TanStack Query	Code	MIT
Drizzle ORM	Code	Apache-2.0
Multer 2	Code	MIT

pdf-parse 2.4.5	Code	Apache-2.0
pdfjs-dist 5.4.296	Code	Apache-2.0
Mammoth 1.13.0	Code	BSD-2-Clause
Tesseract.js 7.0.0	Code	Apache-2.0
@tesseract.js-data/eng 1.0.0	Code	MIT
Qwen 3-4B (Qwen/Qwen3-4B)	Model	Apache-2.0
AI_Job_Market_Dataset.csv (job-title list)	Dataset	CC BY 4.0
Most In-Demand Job Skills of 2026 (Qarera; career_roles.json)	Dataset	CC BY 4.0
DOSM Job Market Insights Q1/Q2 2026 and Dec 2025 Labour Force Survey	Dataset	Malaysia Government Open Data Terms of Use 1.0; attribution included
OpenDOSM Monthly Labour Force Survey (lfs_month)	Dataset	CC BY 4.0
O*NET 31.0 skill ratings	Dataset	CC BY 4.0; attribution and modification notice required
MyMahir Malaysia Critical Occupations List (MyCOL) 2024/2025	Dataset	Malaysia Government Open Data Terms of Use 1.0
Self-made profiles, skill examples	Dataset	None; team-created, no real personal data
Learning Resource and Certification Catalogue	Dataset	Individual provider terms apply; no separate catalogue licence

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24th October 2026